

第9章 控制系统的状态空间分析

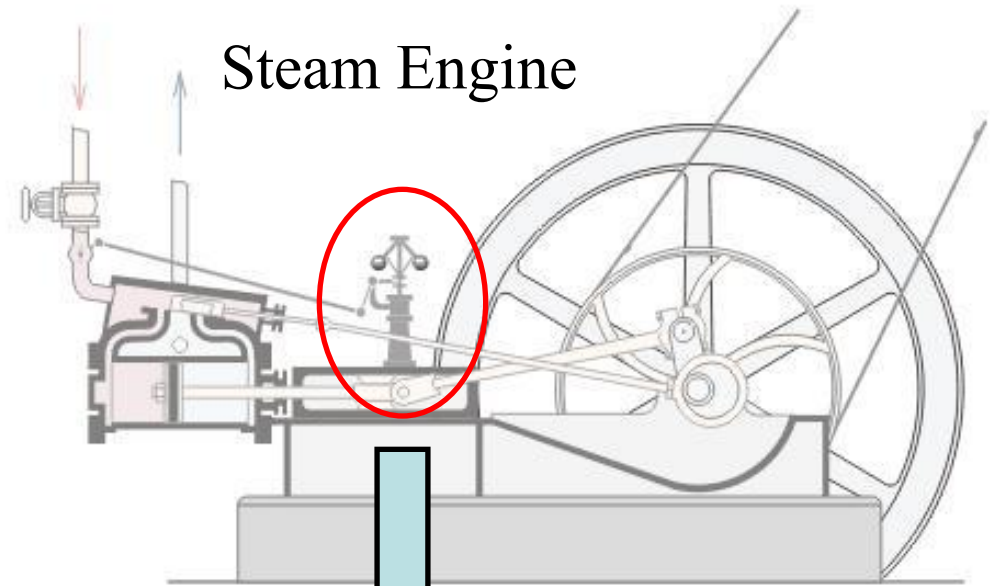
Chapter 9 Analysis of Control Systems in State Space

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- 9.1 Introduction
- 9.2 State-Space and State-Equation
- 9.3 State-Space Establishing of Linear System
- 9.4 Solving the Linear Time-Invariant State Equation
- 9.5 Controllability and Observability
- 9.6 Feedback Structure and State-Observers

- Classify:
 - Classic Control Theory
 - Routh-Hurwitz Stability Criterion, Bode Diagram, Nyquist plot/curve, Nichols plot/curve, Root Locus...
 - Modern Control Theory
 - State Space, Controllability, Observability ...
 - Postmodern control theory
 - Large Scale System, Robust Control, Adaptive Control, Nonlinear Control, Intelligent Control, ...

Mr. J. Watt

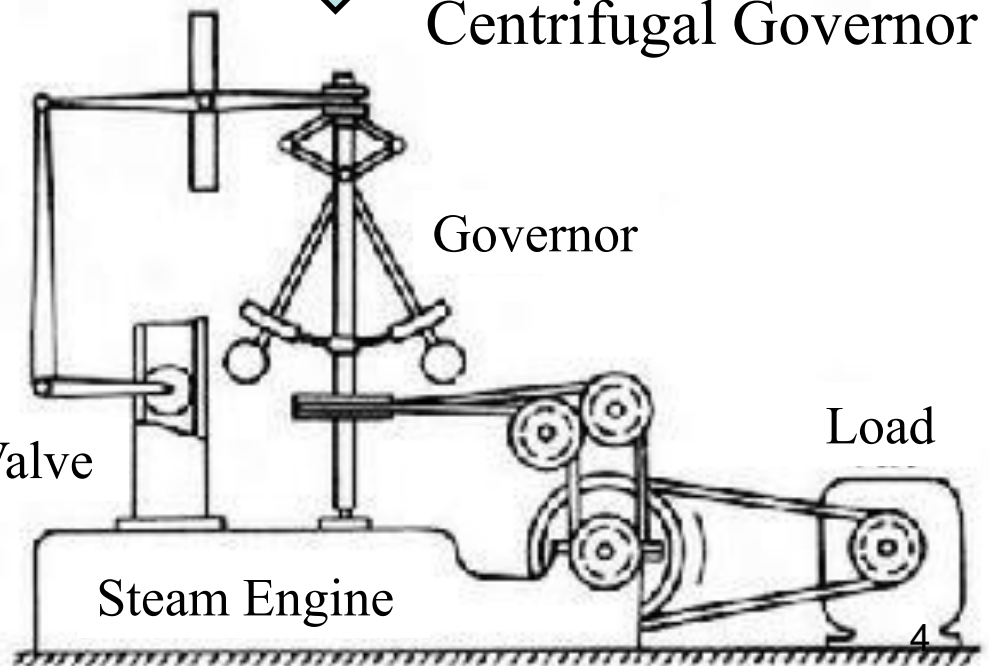


Steam Engine

Centrifugal Governor



Steam Valve

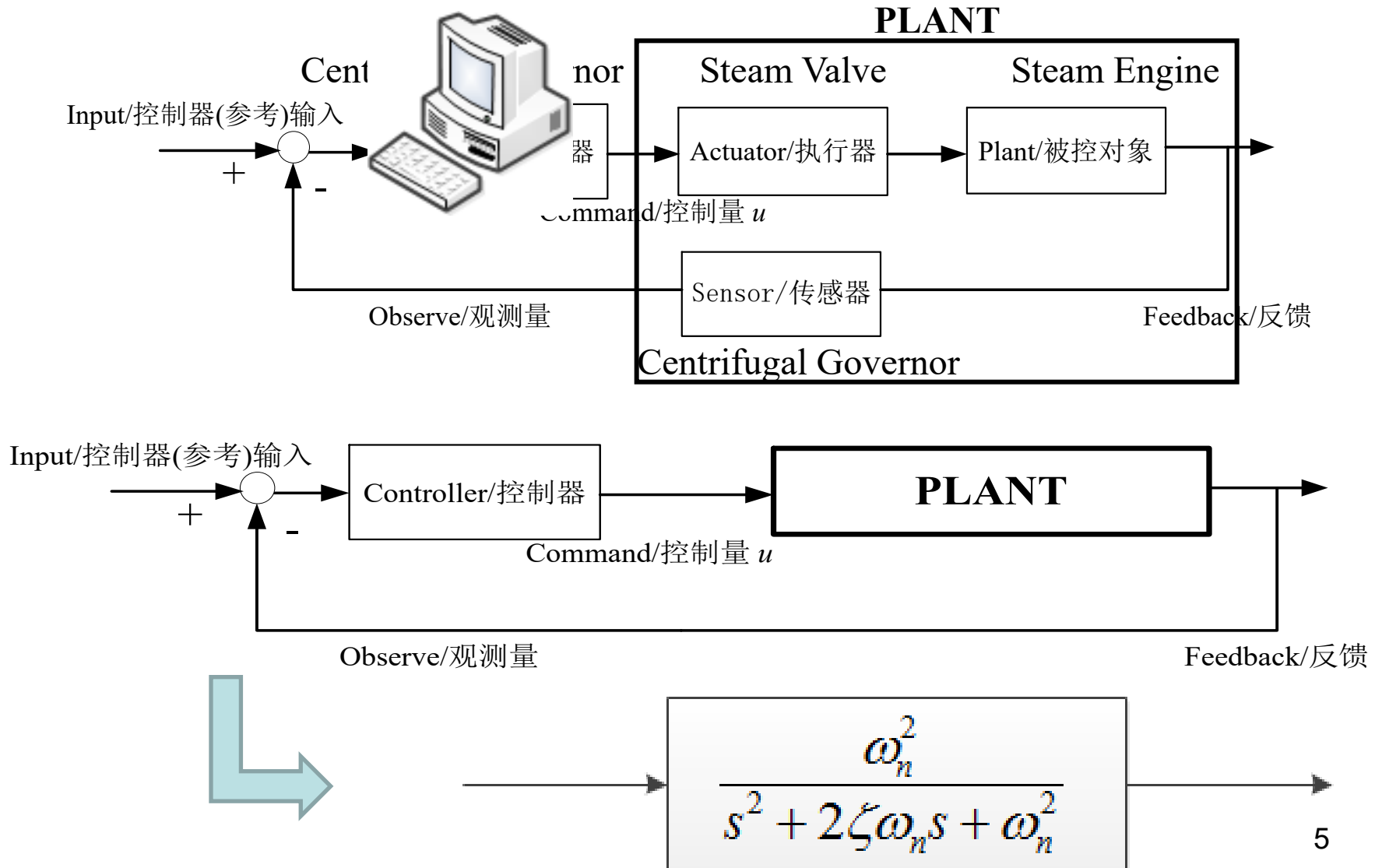


Governor

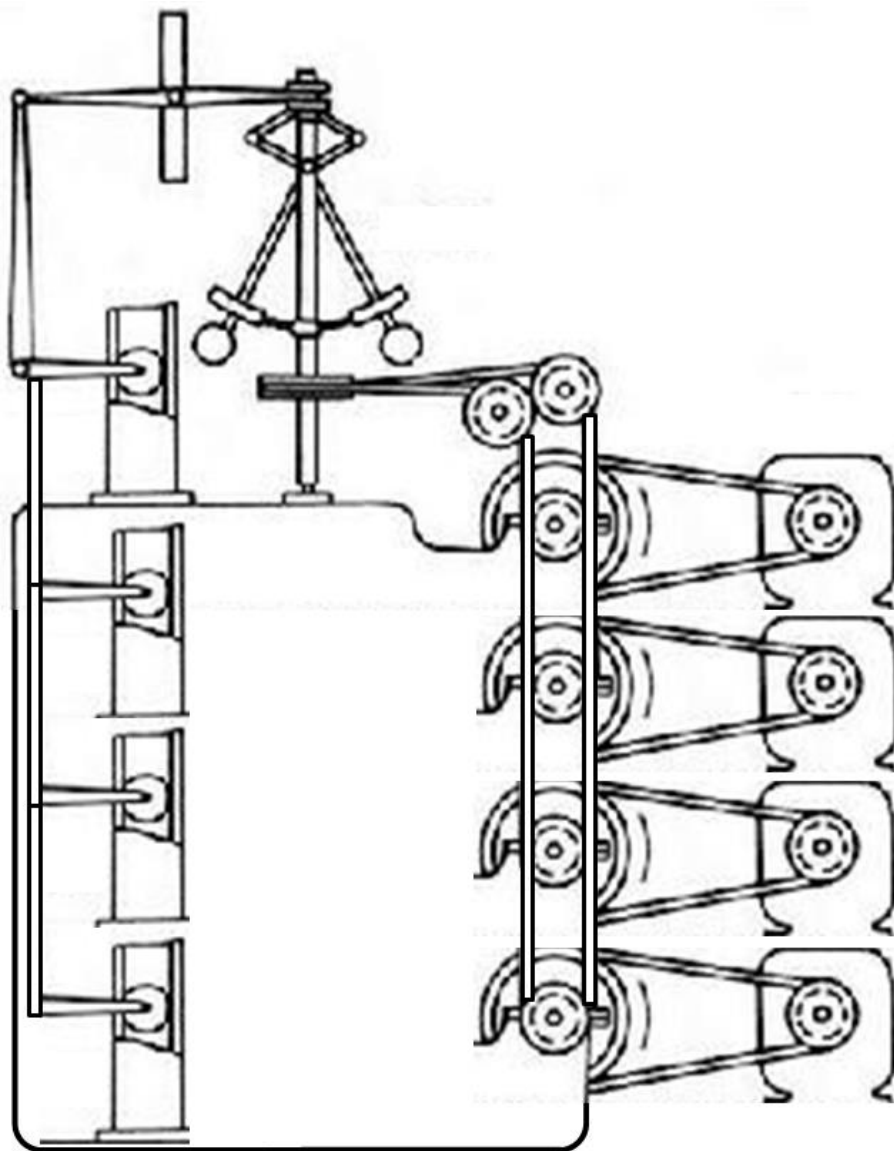
Load

Steam Engine

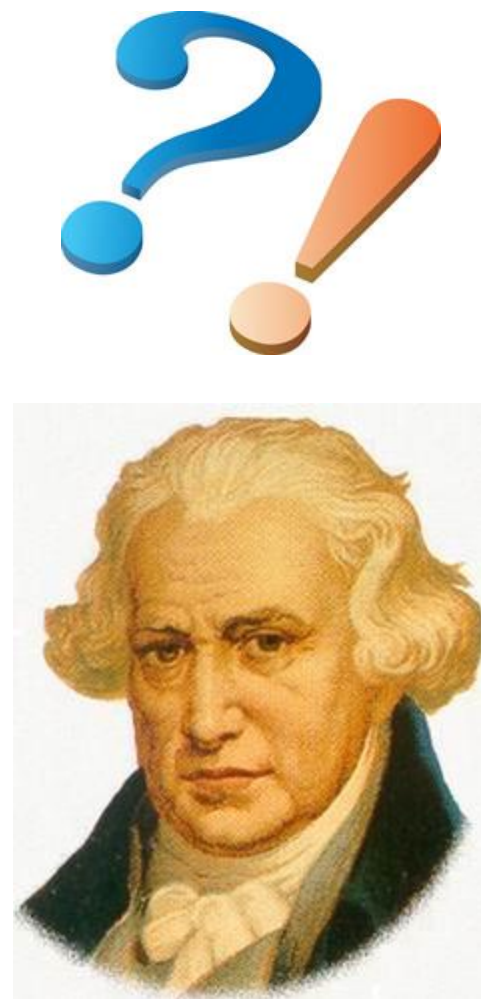
- Review: Control System



n



m



Classic Control Theory

- 1784, Watt, Speed Controller of Steam Engine, Feedback Implement
- 1868, Maxwell, “On Governors” (调节器)
- 1877, Routh stability analysis
- 1895, Hurwitz stability analysis
- 1945, Bode frequent method
- 1948, Evans’ Root Locus
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Classic Control Theory

- Target: SISO, Time-Invariant system
- Description: Differential Equation, Frequency Character, Transfer Function
- Method : Time Domain, Frequency Domain, Root-Locus

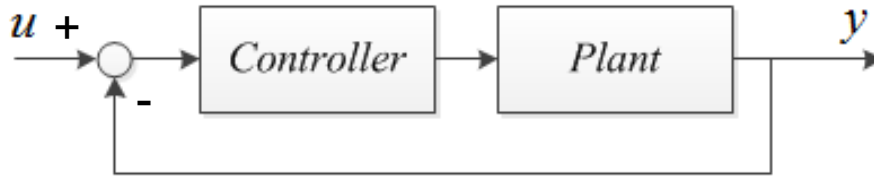
Impactful for Linear, Time-Invariant System Analysis and Design

Limitative for Nonlinear, Time-Variant, Multi-variable System Analysis and Design

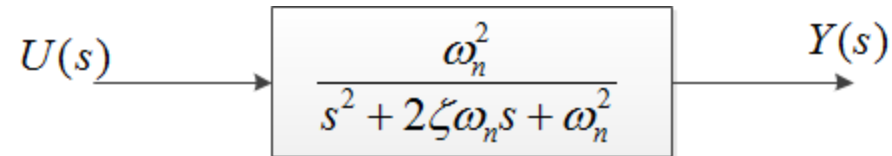
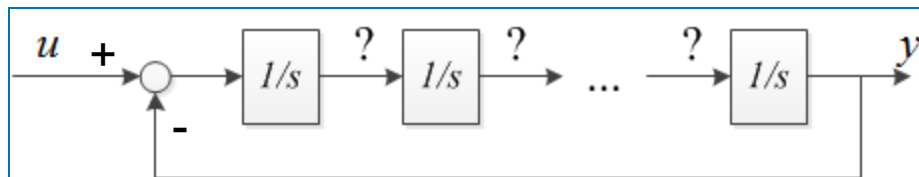
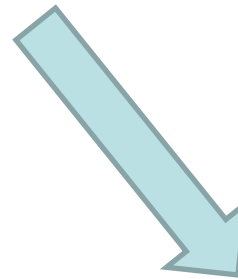
Input / Output relationship, ignore the internal relation of the system, such as coupling(耦合) issue.

Why Modern Control Theory?

- 1. Result and Process

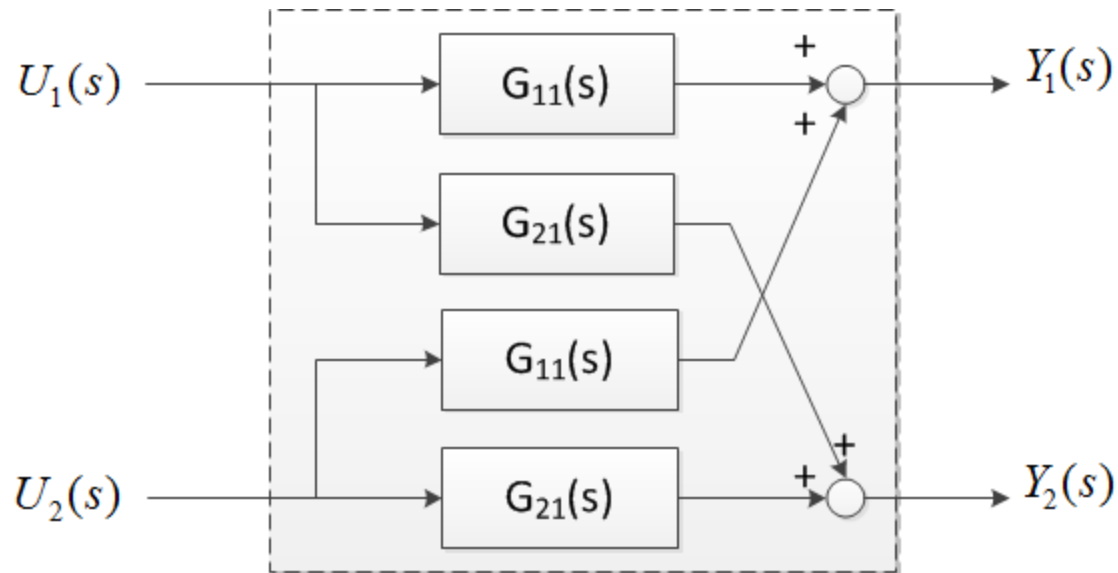


Why feedback?
How feedback?

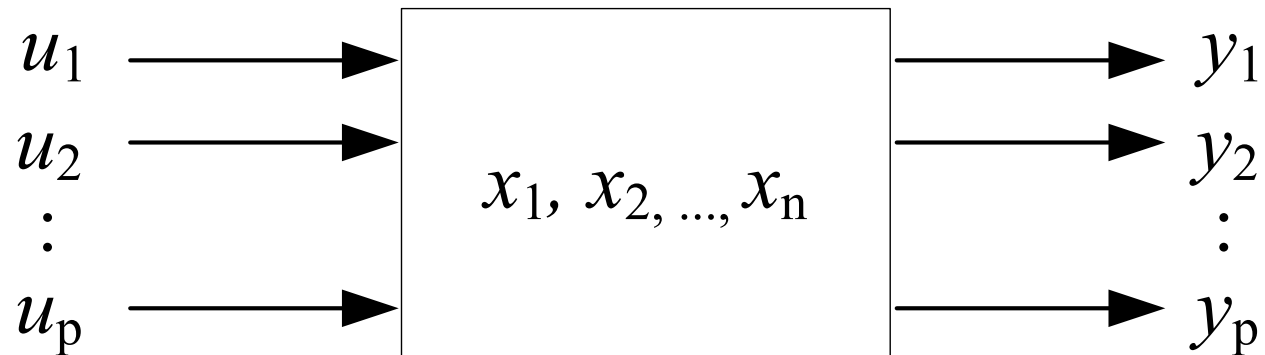


What about the
intermediate states?

- 2. Internal relationship and Coupling(耦合)



- 3. Aerospace science requirement



1950s, “Disordered Time” for Control Theory Development

- Bellman Dynamic Programming / 动态规划
- Pontryagin's Maximum Principle / 极大值原理
- Kalman Controllability & Observability Theory
- Pole Placement / 极点配置
- Observer / 观测器

1960s ~ 1970s, “Modern Control Theory” (State-Space Method)

Modern Control Theory

From Classic to Modern: Kalman proposed State-Space into Control Theory

Fundamental: Linear System Theory, Linear Algebra(线性代数)(Nonlinear Systems and Large-scale Systems are affected by the linear system theory as well), State-Space Analysis

Branches: State-space Method, Linear system Geometrical Theory(线性几何理论), Algebra Theory(代数理论), Multi-variable Frequency Domain Method

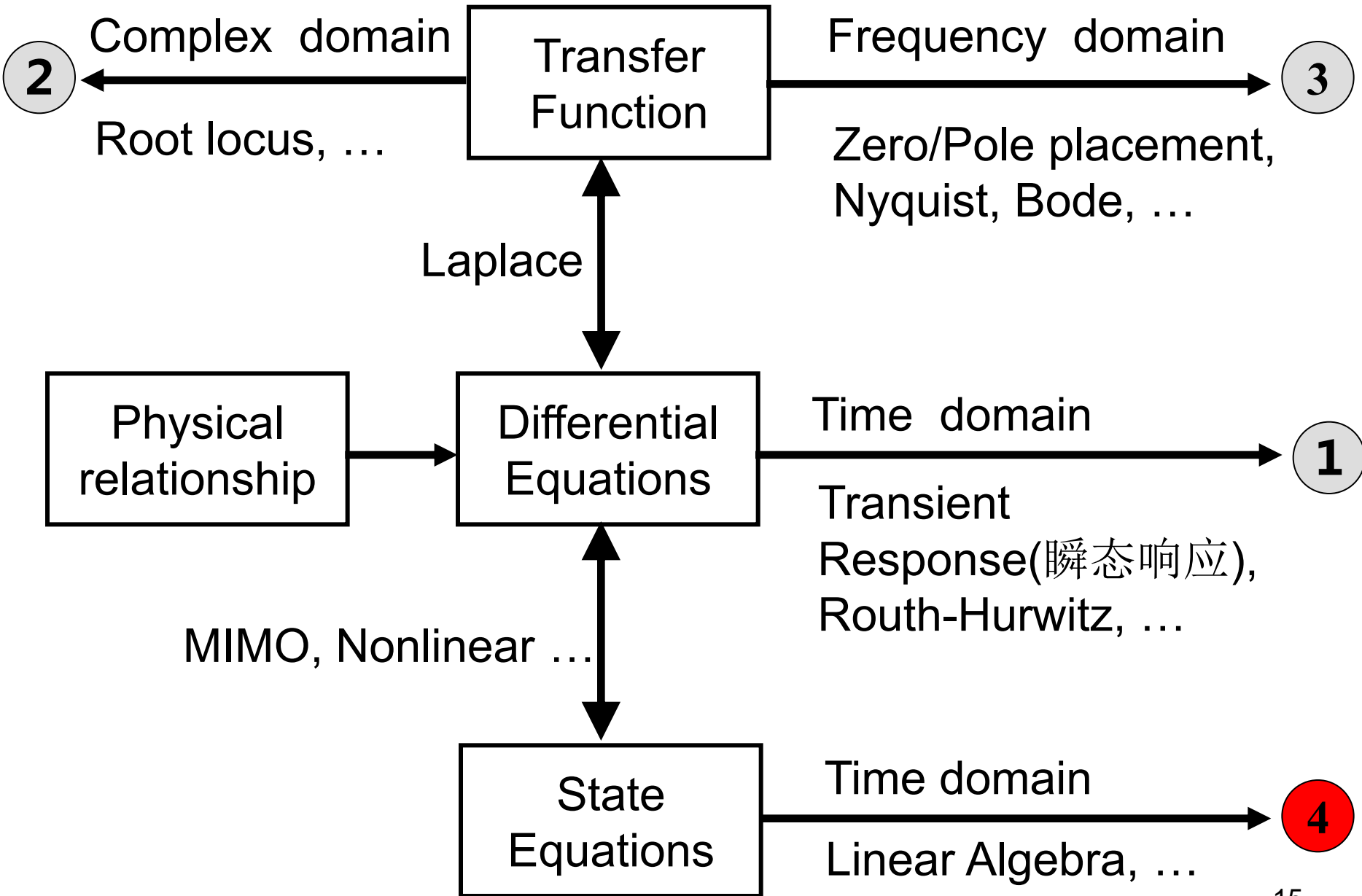
Target: Not only SISO linear time-Invariant Systems, but also MIMO, Nonlinear, time-variant Systems

Areas:

- ❖ Aerospace Science;
- ❖ Robotics;
- ❖ Industry;

Benefit:

- ❖ Mathematical Tools
- ❖ Theory Foundation
- ❖ Research Method
- ❖ External Information(外部信息)(Input&Output Variable)
Internal Information (内部信息)(State Variable)



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9.2 State Space and State Equation

9.2.1 Concepts

1. **State:** In time domain, a set (集合) of variables to describe System's motion and movement information
2. **State-variable: the smallest set** of variables that describe the “state” of a system. Intuitively, the state of a system describes enough about the system to determine its future behavior. In an n^{th} -order differential equation there should be n independent state variables.

Attention:

State-variable is Sufficient and Necessary to describe dynamic motion of the system.(NSC-必要充分条件)

State-variable is not unique

3. **State-vector:** n states variables: $x_1(t), x_2(t), \dots, x_n(t)$ to describe the observed states in State space.

$$X(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$$

4. **State-space:** The n dimension space based on the state-variable $x_1(t), x_2(t), \dots, x_n(t)$.
5. **State-locus:** At special time t_0 , the state $x(t_0)$ will be a point in the state-space; during a period of time t , the state $x(t)$ will be drawn as a trajectory/locus.
6. **State-equation:** The mathematical relationship between two or more state functions, relationship between state variables and the input:

$$\dot{x}(t) = f[x(t), u(t)]$$

$$x(t_{k+1}) = f[x(t_k), u(t_k)]$$

7. **Output-equation:** Describe the relationship between the Output and State, and between Output and Input.

$$y(t) = g[x(t), u(t)]$$

$$y(t_k) = g[x(t_k), u(t_k)]$$

8. **State-space Representation:** Represent the system by the State-equations and Output-equations.

$$\begin{cases} \dot{x}(t) = f[x(t), u(t)] \\ y(t) = g[x(t), u(t)] \end{cases}$$

$$\begin{cases} x(t_{k+1}) = f[x(t_k), u(t_k)] \\ y(t_k) = g[x(t_k), u(t_k)] \end{cases}$$

- State-space Representation

$$\begin{cases} \dot{x}(t) = f[x(t), u(t)] \\ y(t) = g[x(t), u(t)] \end{cases}$$

f and g are linear function, for linear system.

State-equation: a first-order differential equation.

Output-equation: algebraic equation(代数方程) for vectors .

$$\begin{cases} \dot{x}(t) = A(t)x(t) + B(t)u(t) \\ y(t) = C(t)x(t) + D(t)u(t) \end{cases}$$

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

Linear Time-Invariant System

➤ State-equation: Relationship between the Input and State

– For single-input linear time-invariant system

$$\begin{cases} \dot{x}_1(t) = a_{11}x_1(t) + a_{12}x_2(t) + \cdots + a_{1n}x_n(t) + b_1u(t) \\ \dot{x}_2(t) = a_{21}x_1(t) + a_{22}x_2(t) + \cdots + a_{2n}x_n(t) + b_2u(t) \\ \vdots \\ \dot{x}_n(t) = a_{n1}x_1(t) + a_{n2}x_2(t) + \cdots + a_{nn}x_n(t) + b_nu(t) \end{cases}$$

The constant coefficients $a_{11}, \dots, a_{nn}$; $b_1, \dots, b_n$ are decided by system characters.

Matrix expression: $\dot{x}(t) = Ax(t) + bu(t)$

$$x(t) = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \dot{x}(t) = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{pmatrix}, A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ & & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

– For multi-input Linear Time-Invariant System

$$\begin{cases} \dot{x}_1 = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + b_{11}u_1 + b_{12}u_2 + \cdots + b_{1p}u_p \\ \dot{x}_2 = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + b_{21}u_1 + b_{22}u_2 + \cdots + b_{2p}u_p \\ \vdots \\ \dot{x}_n = a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n + b_{n1}u_1 + b_{n2}u_2 + \cdots + b_{np}u_p \end{cases}$$

Matrix expression: $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$

In which

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \dot{\mathbf{x}} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{pmatrix}, \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

$$\mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix}, \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_p \end{pmatrix}$$

➤ Output-equation: relationship between Output & State, and Output & Input.

Output is decided by system task.

- General form Output-equation for single-out linear time-invariant system (单输出线性定常系统)

$$y(t) = c_1 x_1(t) + c_2 x_2(t) + \cdots + c_n x_n(t) + du(t)$$

The constant coefficients $c_1, c_2, \dots, c_n$ and d are relative with the system character.

The Matrix representation:

$$y(t) = \mathbf{c}x(t) + du(t)$$

– General Output function of MIMO system:

$$\left\{ \begin{array}{l} y_1 = c_{11}x_1 + c_{12}x_2 + \cdots + c_{1n}x_n + d_{11}u_1 + d_{12}u_2 + \cdots d_{1p}u_p \\ y_2 = c_{21}x_1 + c_{22}x_2 + \cdots + c_{2n}x_n + d_{21}u_1 + d_{22}u_2 + \cdots d_{2p}u_p \\ \vdots \\ y_q = c_{q1}x_1 + c_{q2}x_2 + \cdots + c_{qn}x_n + d_{q1}u_1 + d_{q2}u_2 + \cdots d_{qp}u_p \end{array} \right.$$

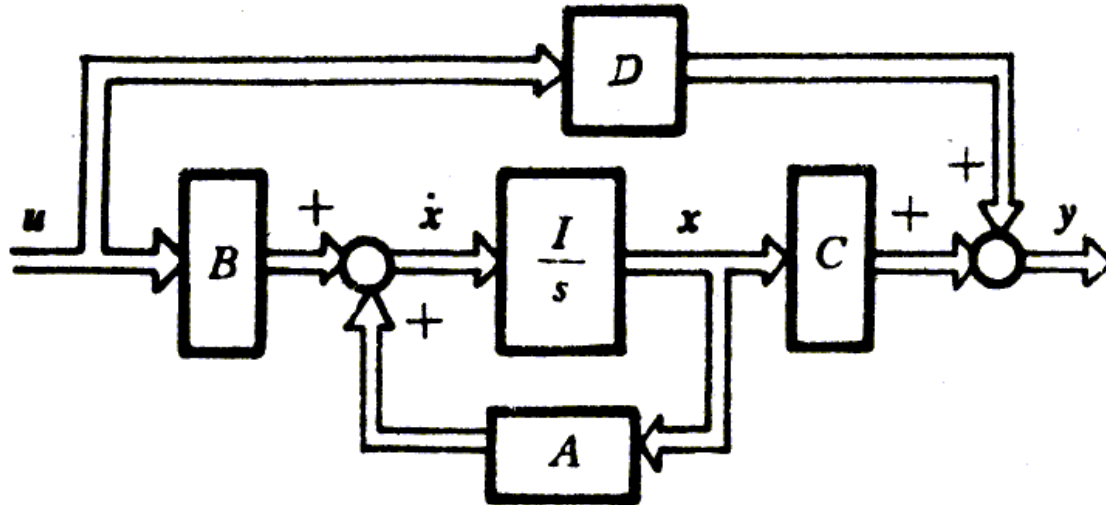
The Matrix representation:

$$y = \mathbf{C}x + \mathbf{D}u$$

State-space equations of Linear Time-Invariant (LTI) System:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$



The Structure of the State-space

A - State Matrix (Systems matrix, coefficients matrix)

B - Input Matrix (Control Matrix)

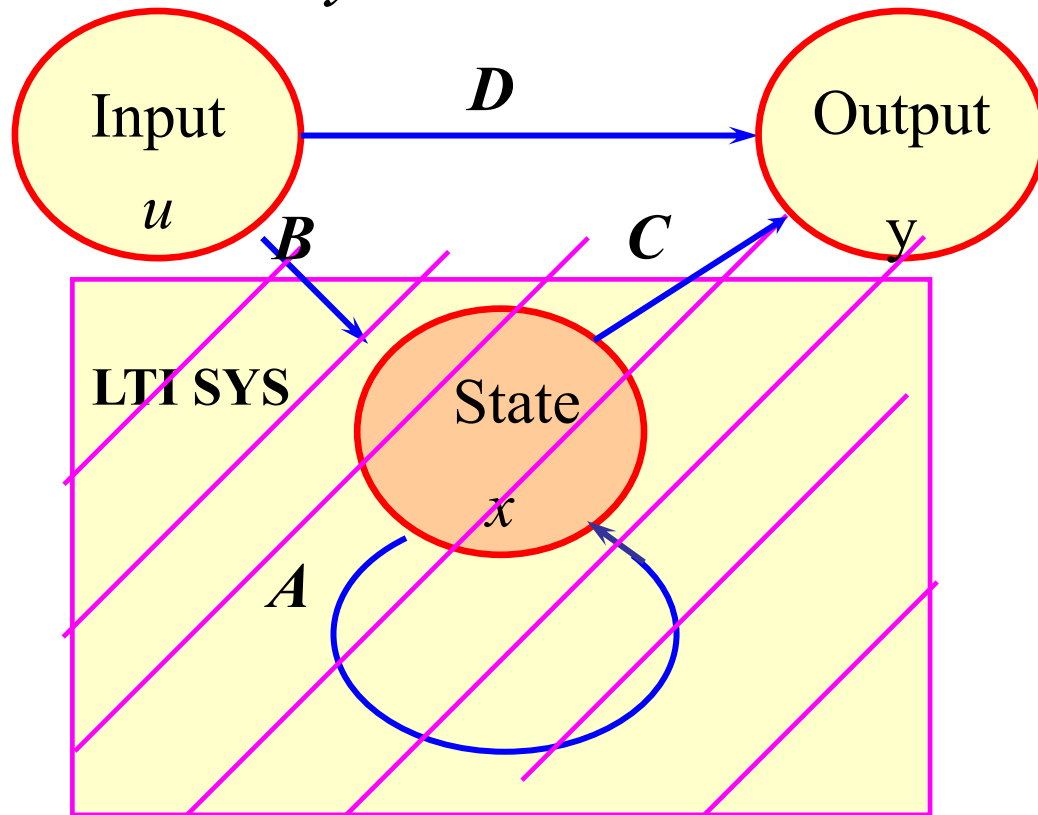
C - Observing Matrix (Output Matrix)

D - Feedforward Matrix (Directly Transfer Matrix)

The LTI system: $\{A, B, C, D\}$

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$



Relationship Chat

Advantage of State-space Analysis Method:

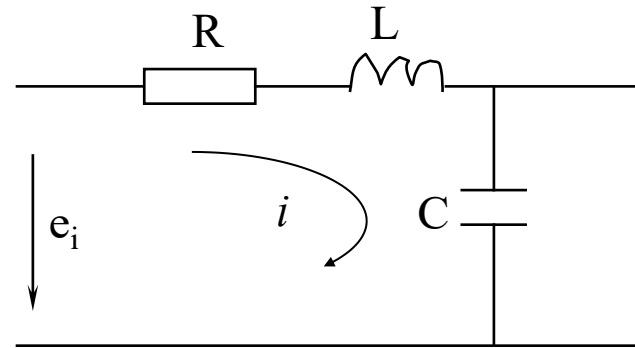
- **Computing:** Using computer to solve the first-order differential equations, easier than higher order equation;
- **Representation:** Using Vector Matrix to simplify the mathematic representation of the differential equations;
- **Field:** MIMO System, Time-Invariant System, Stochastic Process and Sample System, etc.
- **Special:** the use of the state space representation is not limited to systems with linear components and zero initial conditions.

9.2.2 Examples

Ex. 9-1 RLC electric circuit 's State-space model (figure).

Physics relationship:

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = e_i$$



If assume $e_i(t)$ is the input: $u(t)$, $i(t)$ is the output: $y(t)$, and select the proper state-variables $i(t)$ and $\int i(t)dt$:

$$\begin{aligned} x_1(t) &= i(t) \\ x_2(t) &= \int i(t)dt \end{aligned} \quad \Rightarrow \quad \begin{aligned} \frac{dx_1(t)}{dt} &= -\frac{R}{L} x_1(t) - \frac{1}{LC} x_2(t) + \frac{1}{L} u(t) \\ \frac{dx_2(t)}{dt} &= x_1(t) \end{aligned}$$

$$\frac{dx_1(t)}{dt} = -\frac{R}{L}x_1(t) - \frac{1}{LC}x_2(t) + \frac{1}{L}u(t)$$

$$\frac{dx_2(t)}{dt} = x_1(t)$$

Rewrite to State equations:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{LC} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u(t)$$

A

B

$y(t)=i(t)=x_1(t)$, thus we have Output equation:

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

C

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t)$$

$$y(t) = C\mathbf{x}(t)$$

Not the end!!!!

Furthermore!!!!

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = e_i$$

If select some another state variables:

$$x_1 = \frac{1}{C} \int i dt + Ri, x_2 = \frac{1}{C} \int i dt$$

then

$$x_1 = x_2 + Ri, \quad L \frac{di}{dt} + x_1 = e_i$$

we have

$$\begin{cases} \dot{x}_1 = \dot{x}_2 + R \frac{di}{dt} = \frac{1}{RC} (x_1 - x_2) + \frac{R}{L} (-x_1 + e_i) \\ \dot{x}_2 = \frac{1}{C} i = \frac{1}{RC} (x_1 - x_2) \\ y = \frac{1}{R} x_1 - \frac{1}{R} x_2 \end{cases}$$

**State-space
representation:**

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{RL} - \frac{R}{L} & -\frac{1}{RC} \\ \frac{1}{RC} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{R}{L} \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} \frac{1}{R} & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Comparison:

$$A = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{LC} \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} \frac{1}{RL} - \frac{R}{L} & -\frac{1}{RC} \\ \frac{1}{RC} & -\frac{1}{RC} \end{bmatrix} \quad B = \begin{bmatrix} \frac{R}{L} \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} \frac{1}{R} & -\frac{1}{R} \end{bmatrix}$$

Conclusion: State-space is non-unique.

- ❖ State-variable is not unique. Different state-equations with
- ❖ State-equation is not unique. Different state-variable

1st selection: $x_1(t) = i(t)$

$$x_2(t) = \int i(t) dt$$

2nd selection: $\bar{x}_1 = \frac{1}{c} \int i dt + Ri$

$$\bar{x}_2 = \frac{1}{c} \int i dt$$

we have

$$x_1 = \frac{1}{R} \bar{x}_1 - \frac{1}{R} \bar{x}_2$$

$$x_2 = c \bar{x}_2$$

matrix representation:

$$\mathbf{x} = \mathbf{P} \bar{\mathbf{x}} \quad \text{in which} \quad \mathbf{P} = \begin{bmatrix} \frac{1}{R} & -\frac{1}{R} \\ 0 & c \end{bmatrix}$$

Infinite groups of state-variables $\mathbf{x}$ are available by any different nonsingular matrix (非奇异阵) $\mathbf{P}$.

